```
Uganda, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -39567.7            -39555.3                 0
     2 20263017             5000           0.277             -39624.0            -39614.7                 0
     3 20263018             5000           0.277             -39602.3            -39593.0                 0
     4 20263019             5000           0.276             -39607.2            -39603.9                 0
                                    file
 mcmc_uganda_step2_chain1_2026-10-31.Rdata
 mcmc_uganda_step2_chain2_2026-10-31.Rdata
 mcmc_uganda_step2_chain3_2026-11-01.Rdata
 mcmc_uganda_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
  age.65.to.79.mortality.slope.multiplier.male         age.30.to.39.transmission.multiplier.3
                                         3.239                                          2.682
                 hiv.specific.mortality.rates.2                male.hiv.mortality.multiplier.0
                                         2.353                                          2.247
        age.40.to.49.transmission.multiplier.2         age.15.to.19.transmission.multiplier.3
                                         2.194                                          2.132
                                       trate.2         age.20.to.29.transmission.multiplier.0
                                         2.095                                          2.063
                               age.assortativity             over.50.hiv.mortality.multiplier.2
                                         1.981                                          1.941
        age.15.to.19.transmission.multiplier.2                                        trate.3
                                         1.852                                          1.822
        age.0.to.14.hiv.mortality.multiplier.2         age.15.to.19.transmission.multiplier.0
                                         1.783                                          1.655
 over.80.mortality.intercept.multiplier.female         age.20.to.29.transmission.multiplier.1
                                         1.644                                          1.588
                                       trate.0      over.80.mortality.slope.multiplier.female
                                         1.586                                          1.553
        age.40.to.49.transmission.multiplier.0             over.50.hiv.mortality.multiplier.0
                                         1.533                                          1.488
                 hiv.specific.mortality.rates.0   age.45.to.64.mortality.slope.multiplier.male
                                         1.455                                          1.443
                                       trate.4        age.15.to.24.hiv.mortality.multiplier.0
                                         1.428                                          1.424
age.65.to.79.mortality.slope.multiplier.female
                                         1.370

Merged simset:
 location simulations total.weight                                        file
   Uganda        1000         1000 simset_uganda_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 56 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 female              26                   0              x0.26      2025            x0.02
 hiv.mortality 50 and over female              26                  12              x3.81      2025            x3.77
 hiv.mortality       25-49   both              26                   0              x0.34      2025            x0.02
 hiv.mortality 50 and over   both              26                   8              x2.87      2025            x2.59
 hiv.mortality       25-49   male              26                   0              x0.41      2025            x0.01
     incidence       10-19   male              26                   0              x2.39      2025            x2.97
 hiv.mortality 50 and over   male              26                   8              x2.16      2025            x1.76
   suppression        0-14   both              10                   0          -20.1 pts      2025        -6.1 pts
    engagement        0-14   both              11                   0          -18.5 pts      2025         -3.2 pts
```

Uganda step 2: likelihood

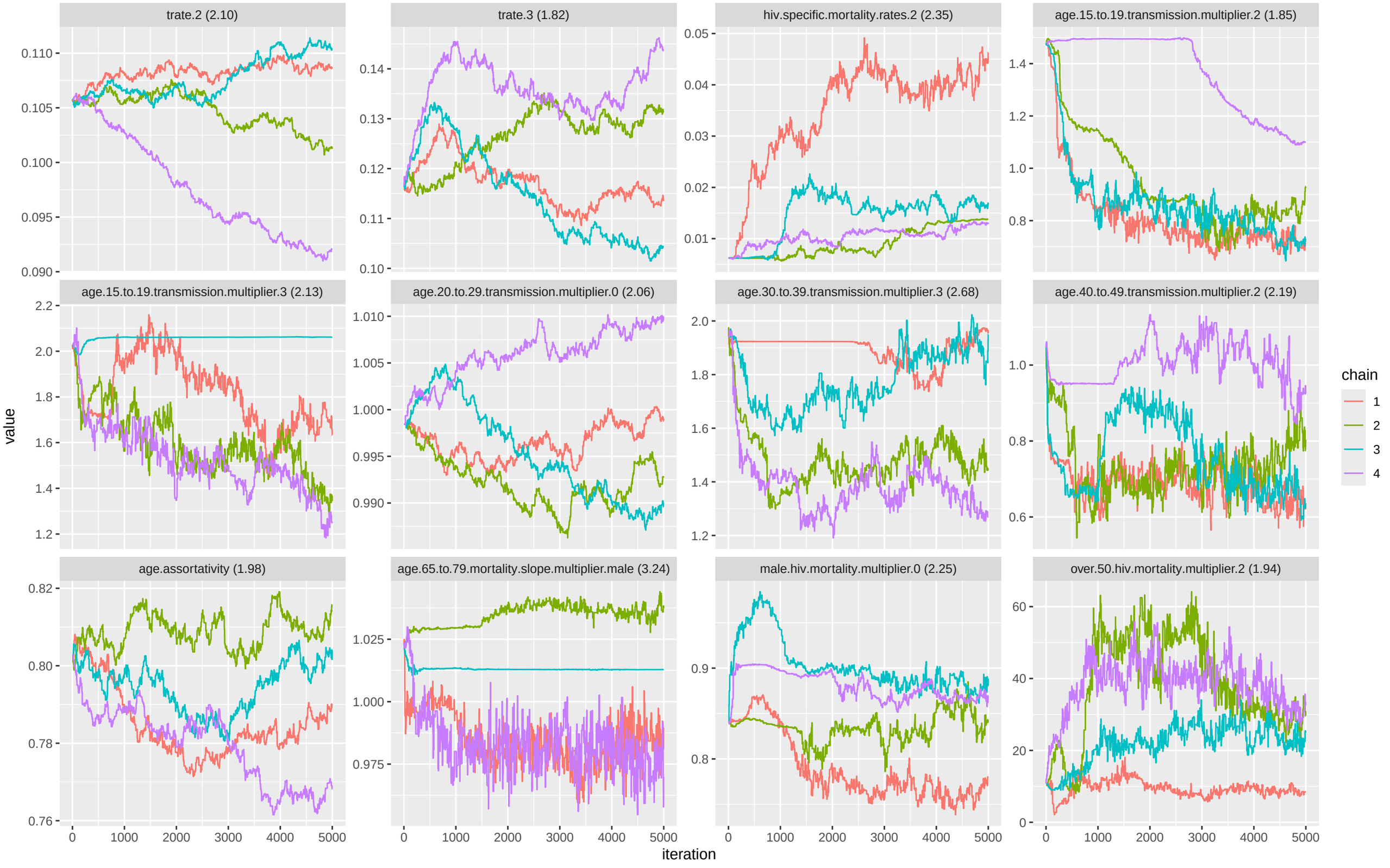
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

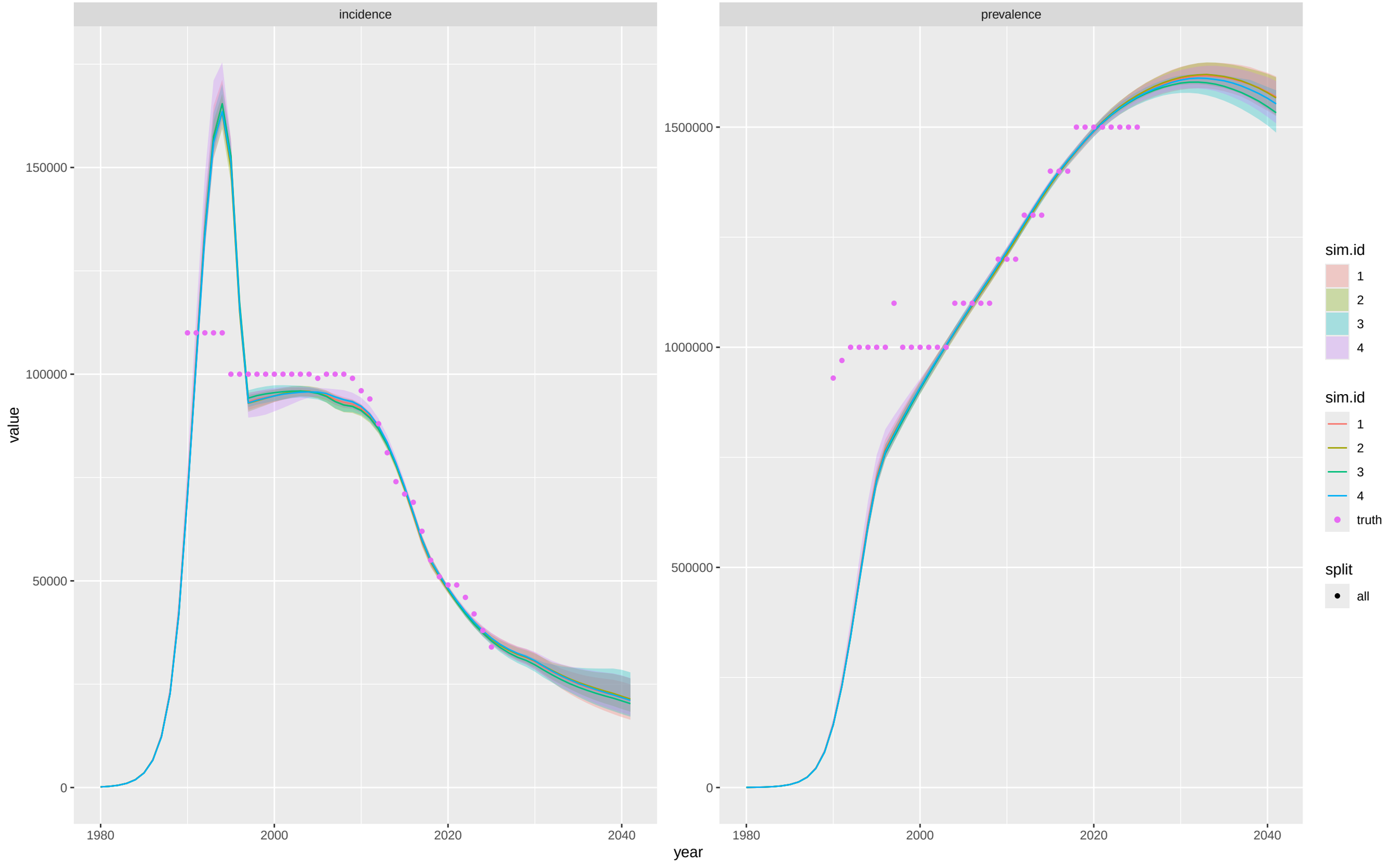
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

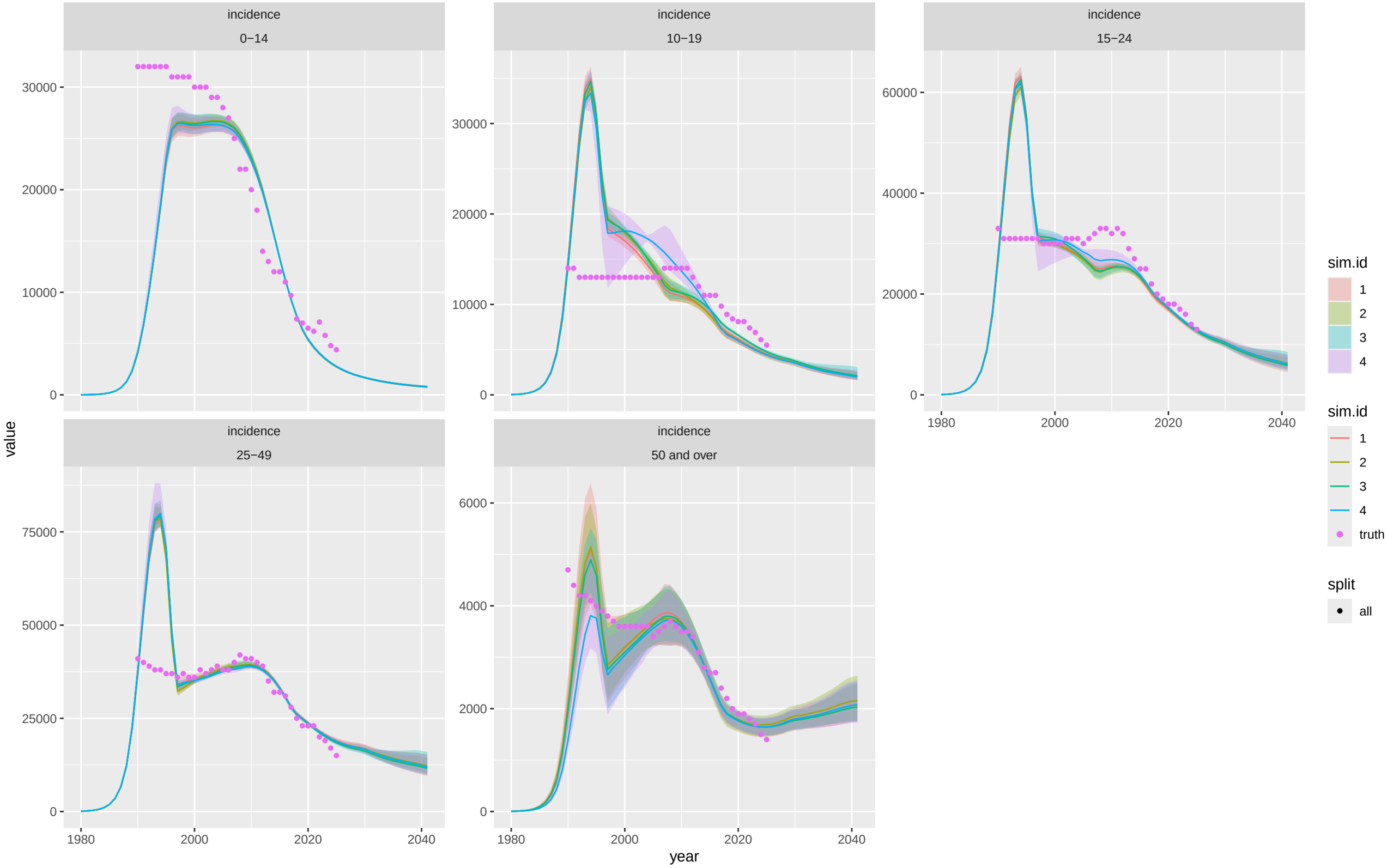
Uganda step 2: trace



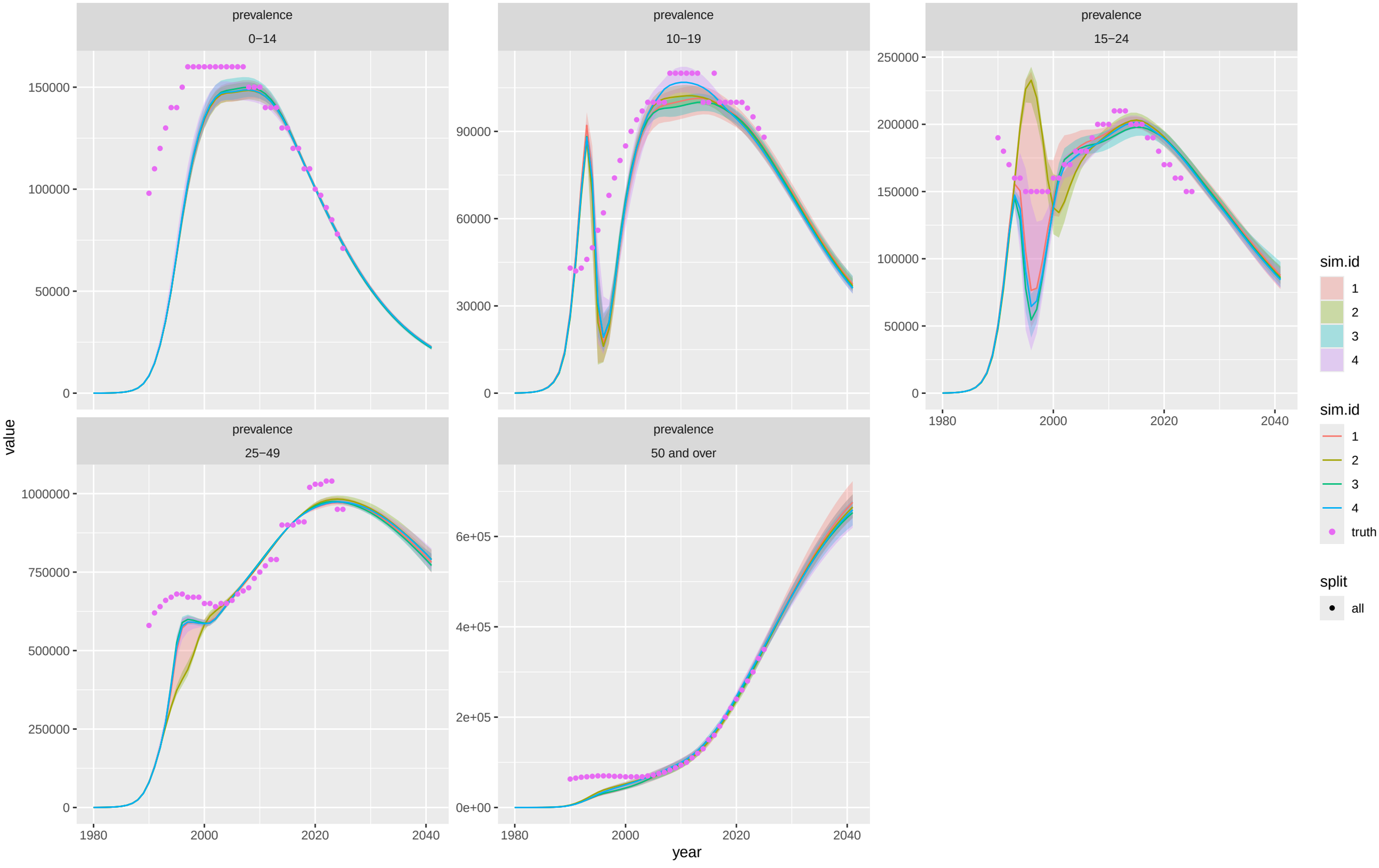
Uganda step 2: fit (one color per chain)



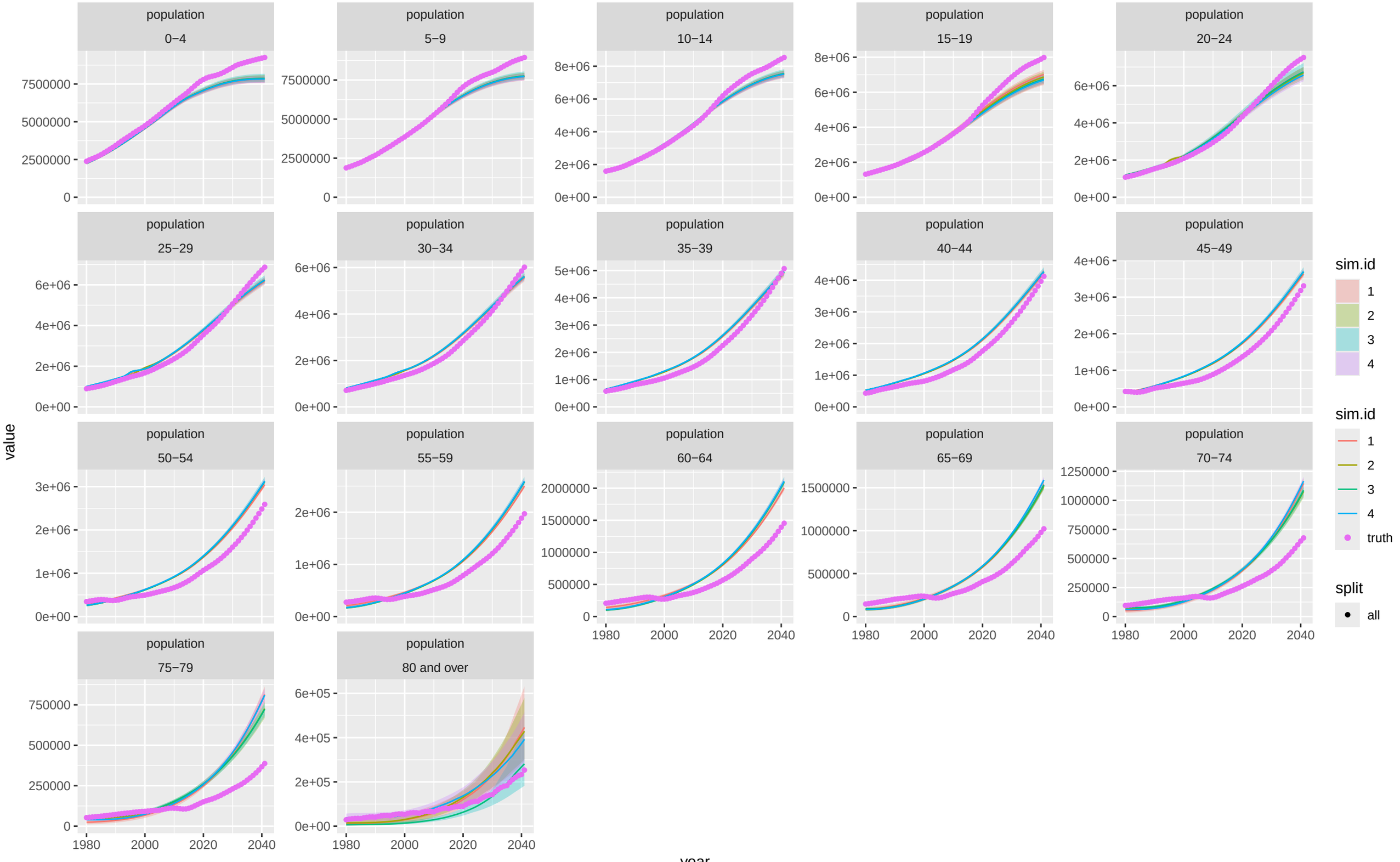
Uganda step 2: incidence.by.age (one color per chain)



Uganda step 2: prevalence.by.age (one color per chain)



Uganda step 2: population.by.age (one color per chain)



Uganda step 2: cascade (one color per chain)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

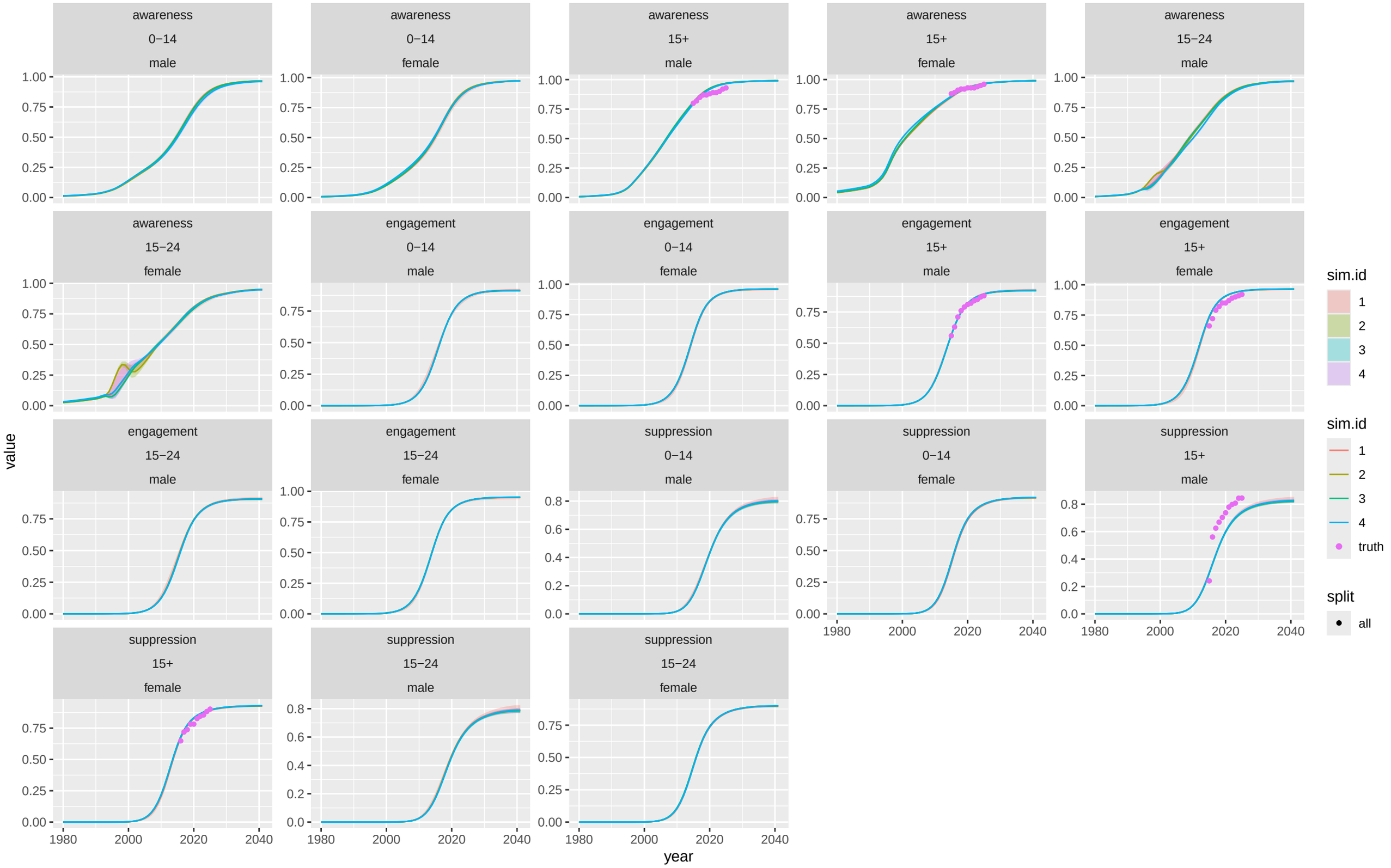
1980
2000
2020
2040

sim.id
1
2
3
4

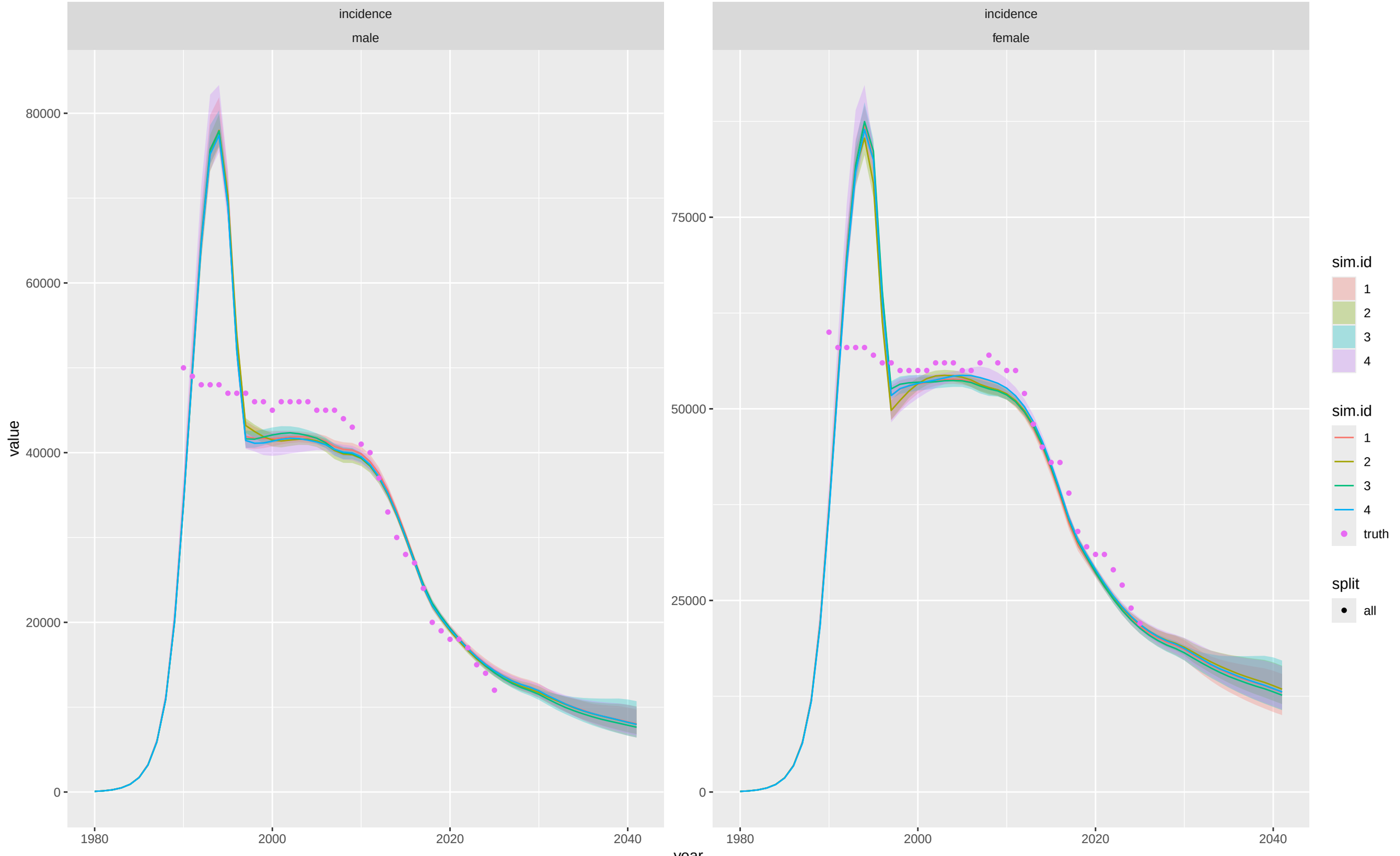
sim.id
1
2
3
4
truth

split
all

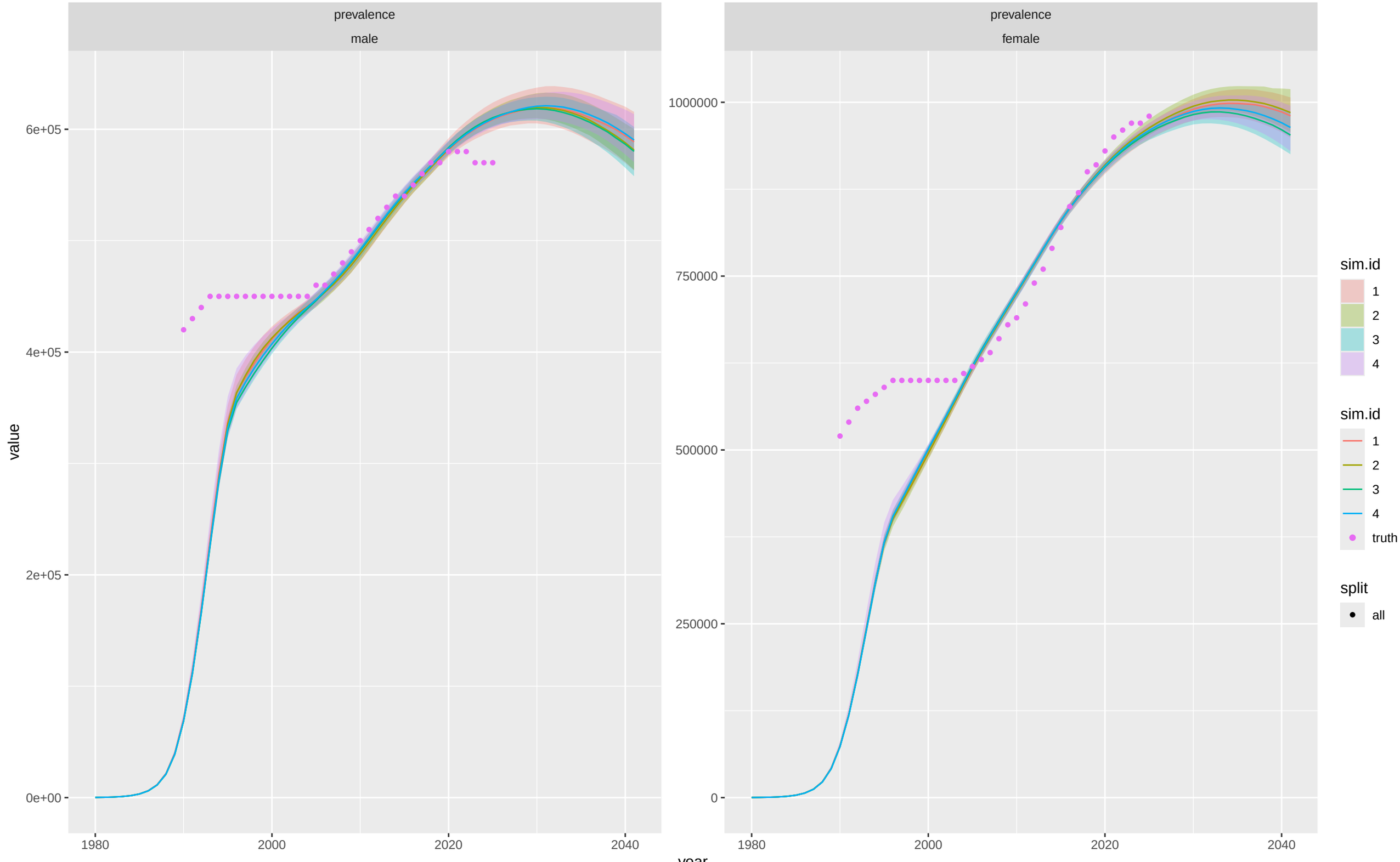
Uganda step 2: cascade.by.age.sex (one color per chain)



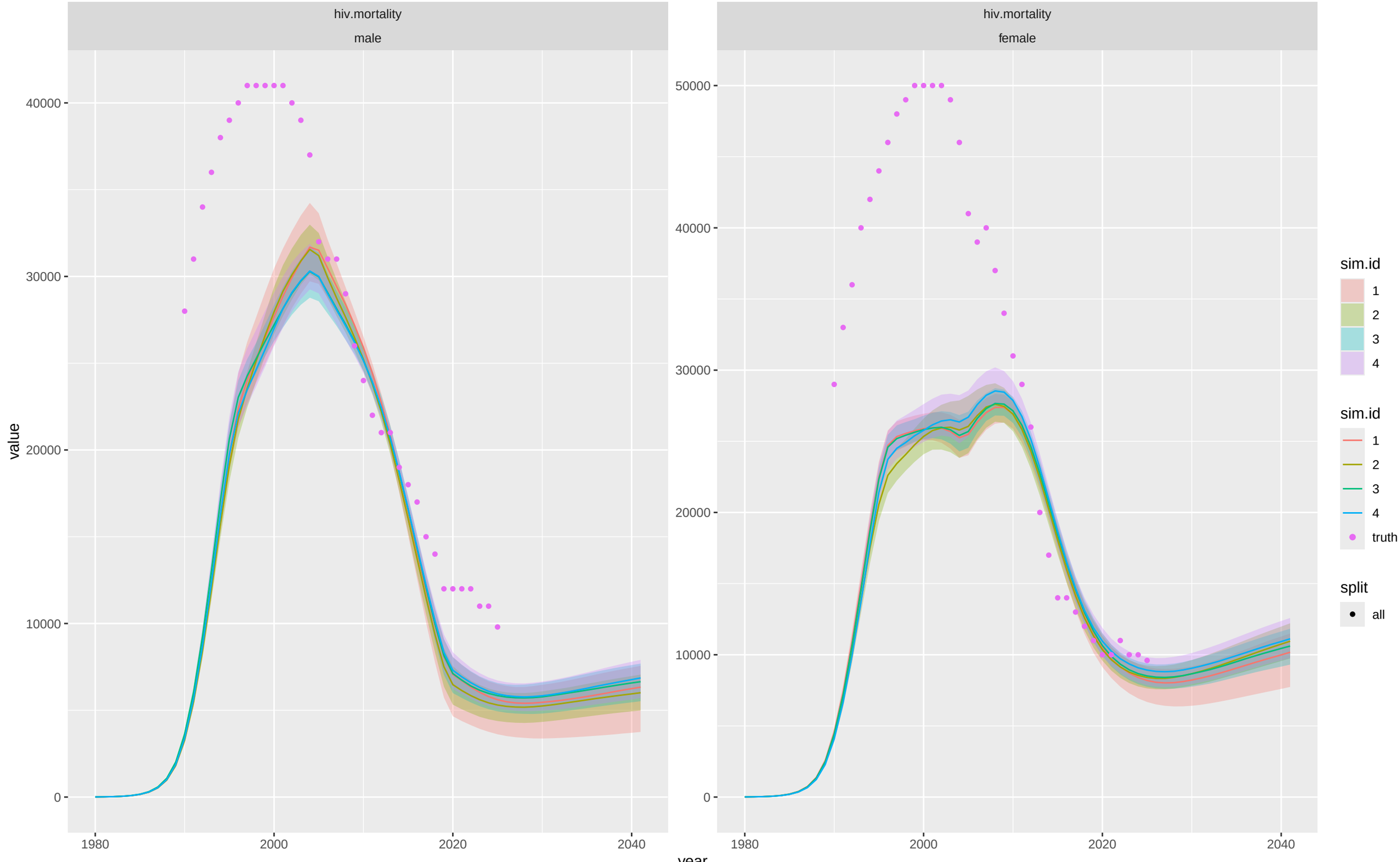
Uganda step 2: incidence.by.sex (one color per chain)



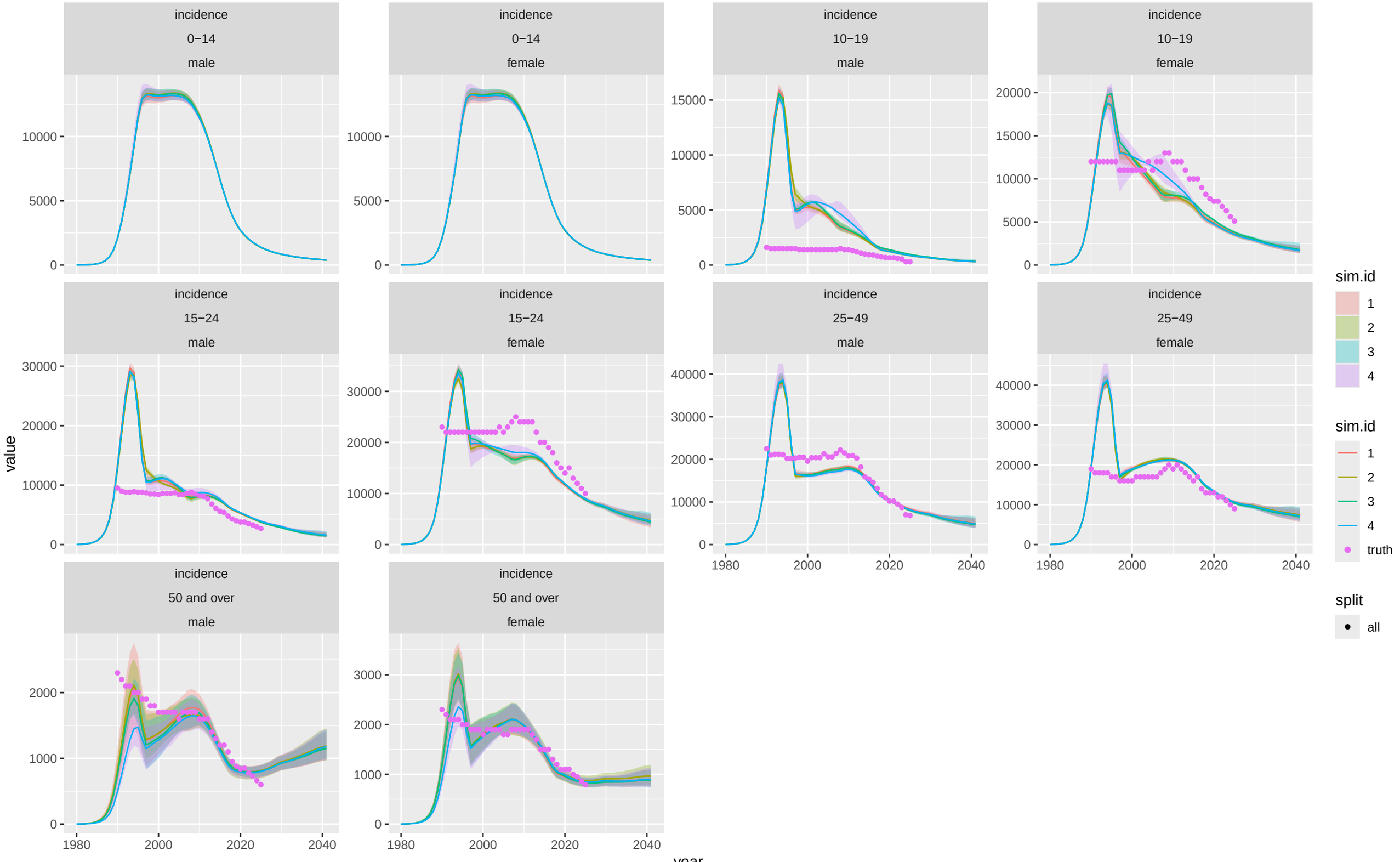
Uganda step 2: prevalence.by.sex (one color per chain)



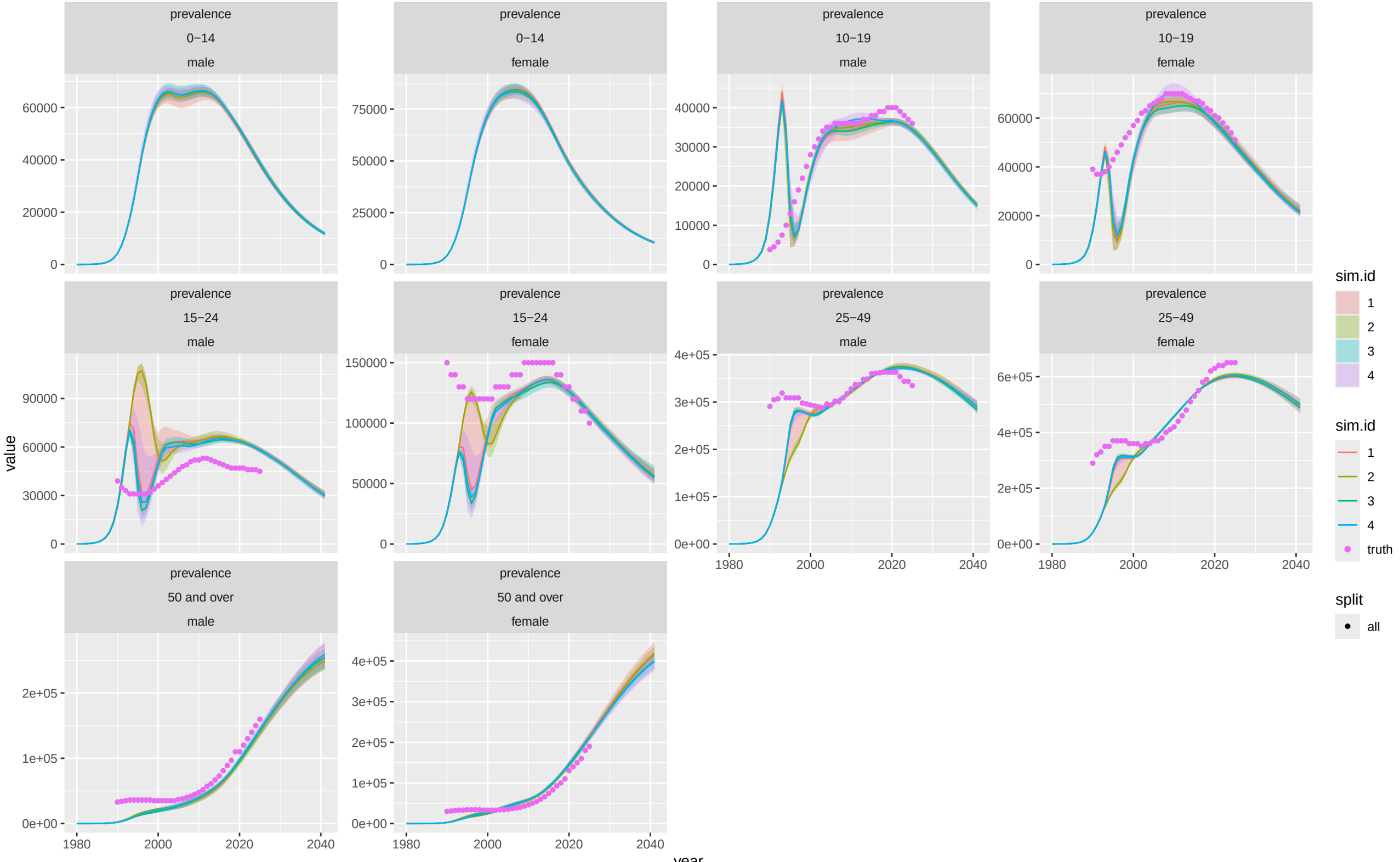
Uganda step 2: hiv.mortality.by.sex (one color per chain)



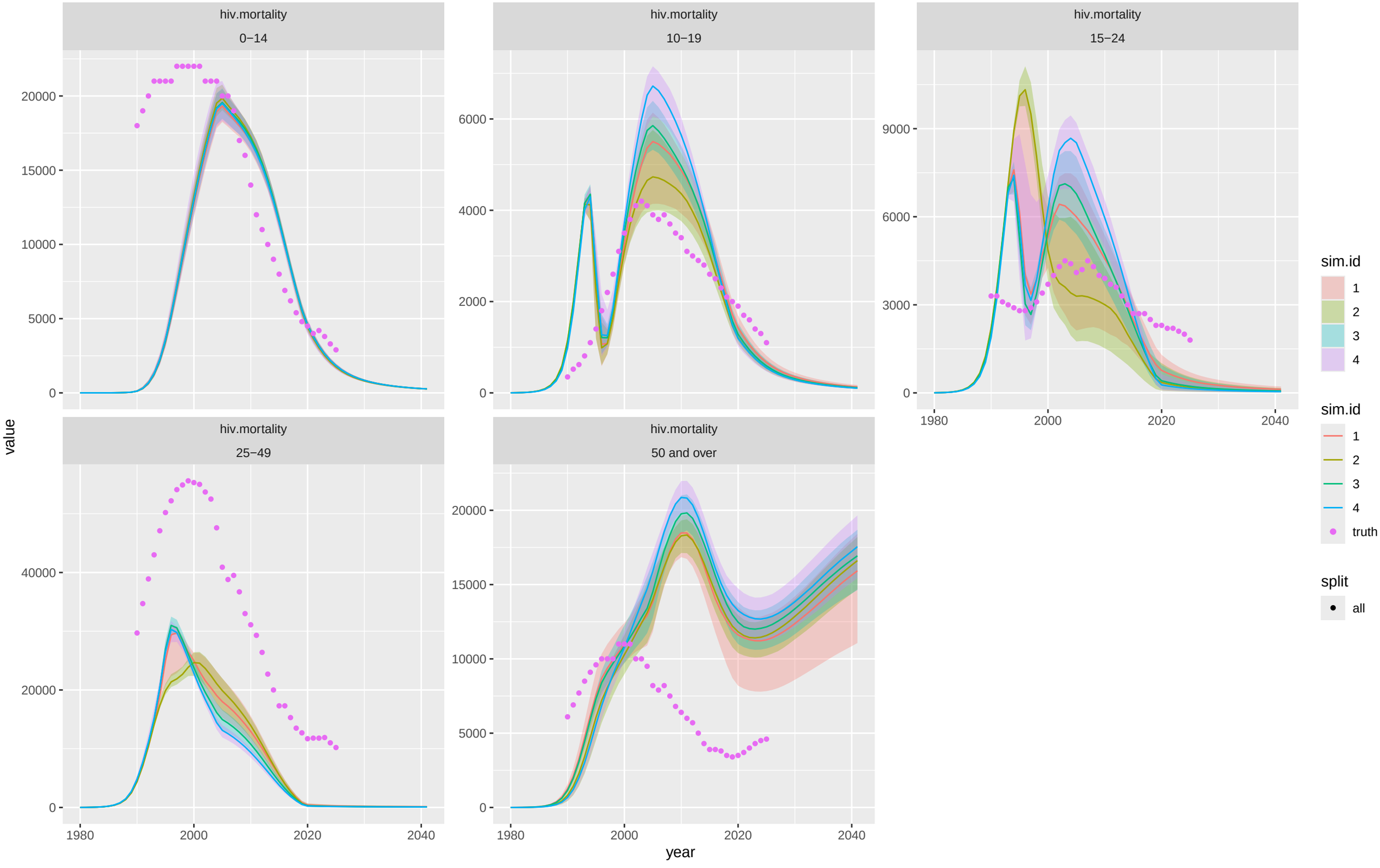
Uganda step 2: incidence.by.age.sex (one color per chain)



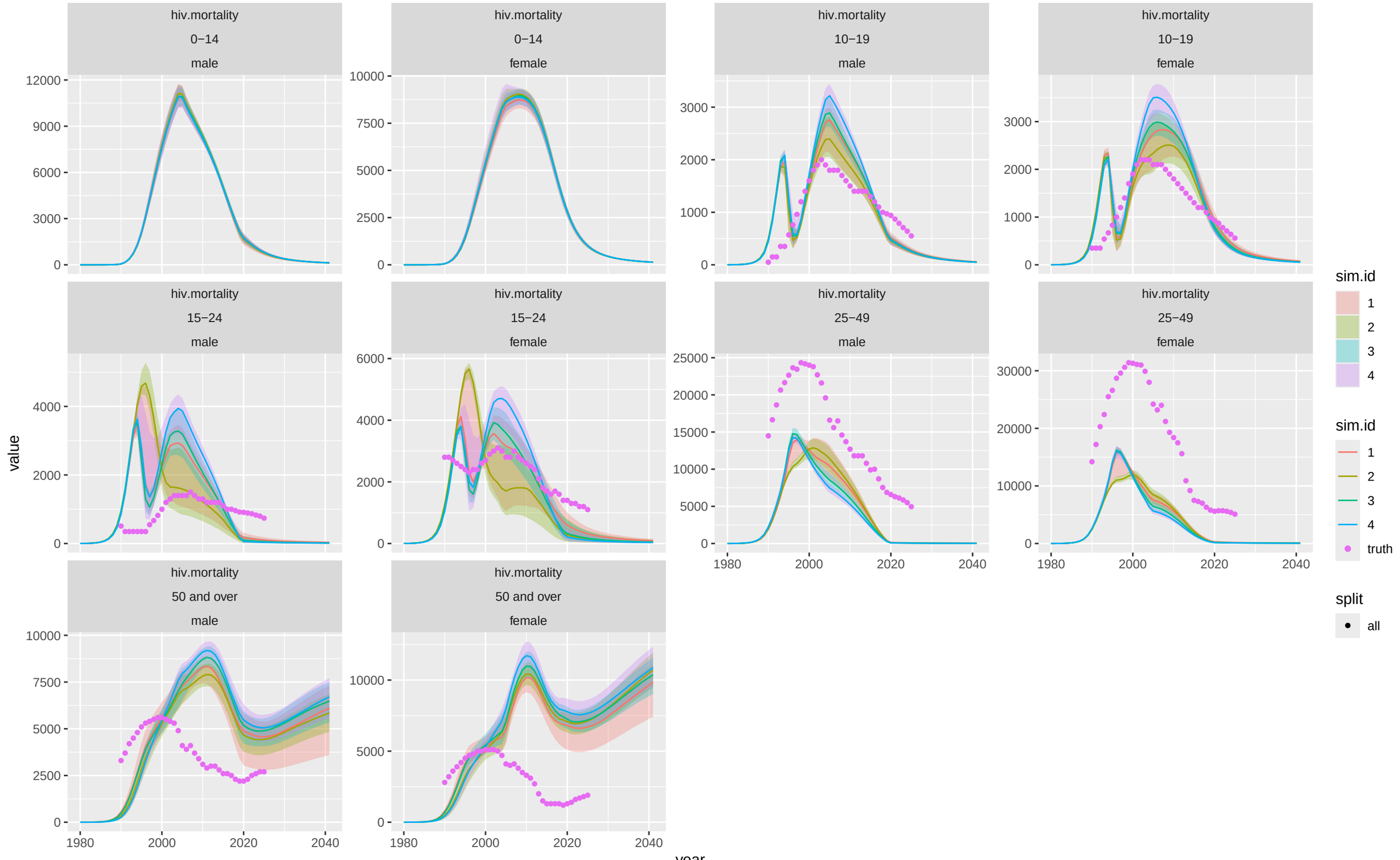
Uganda step 2: prevalence.by.age.sex (one color per chain)



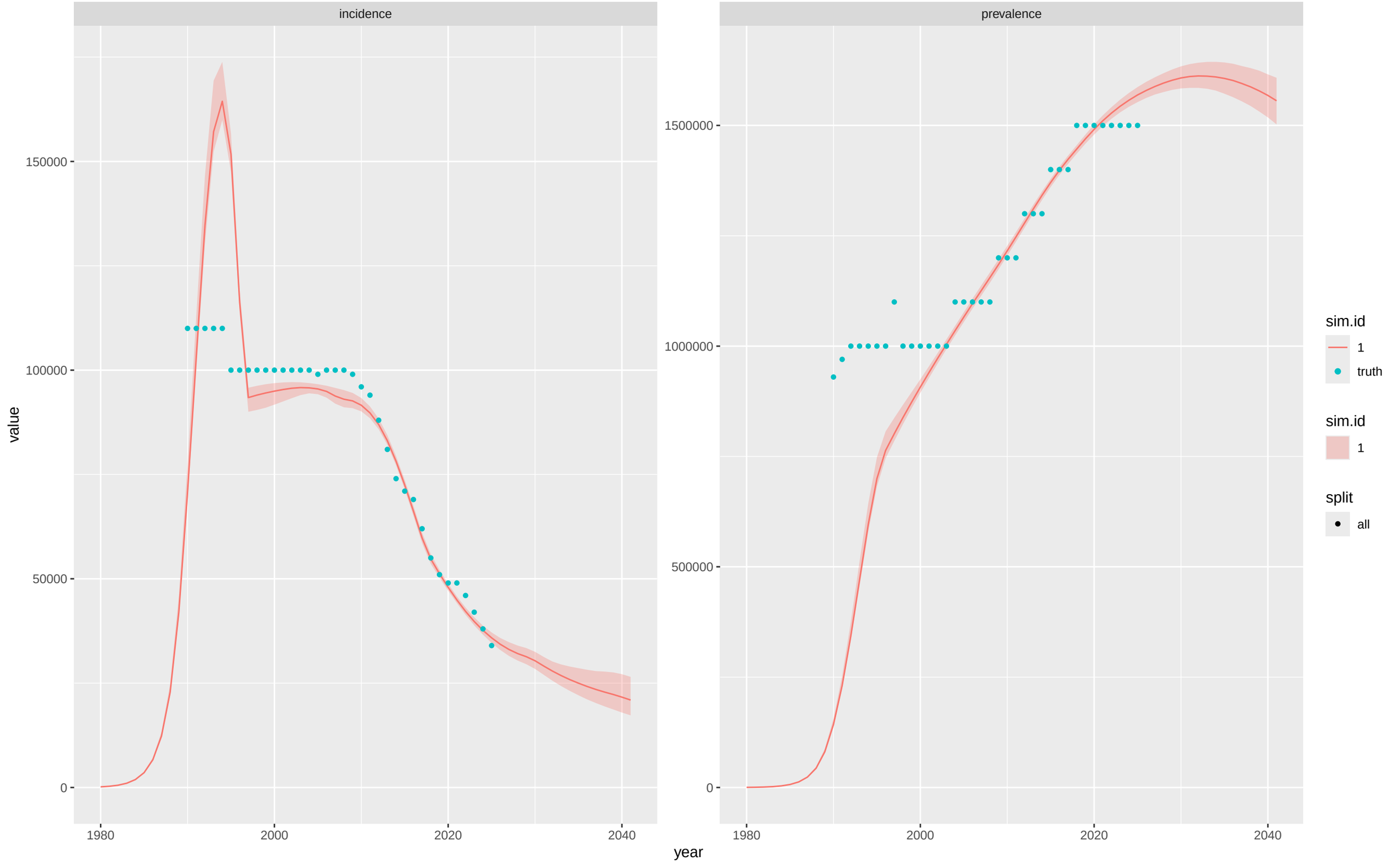
Uganda step 2: hiv.mortality.by.age (one color per chain)



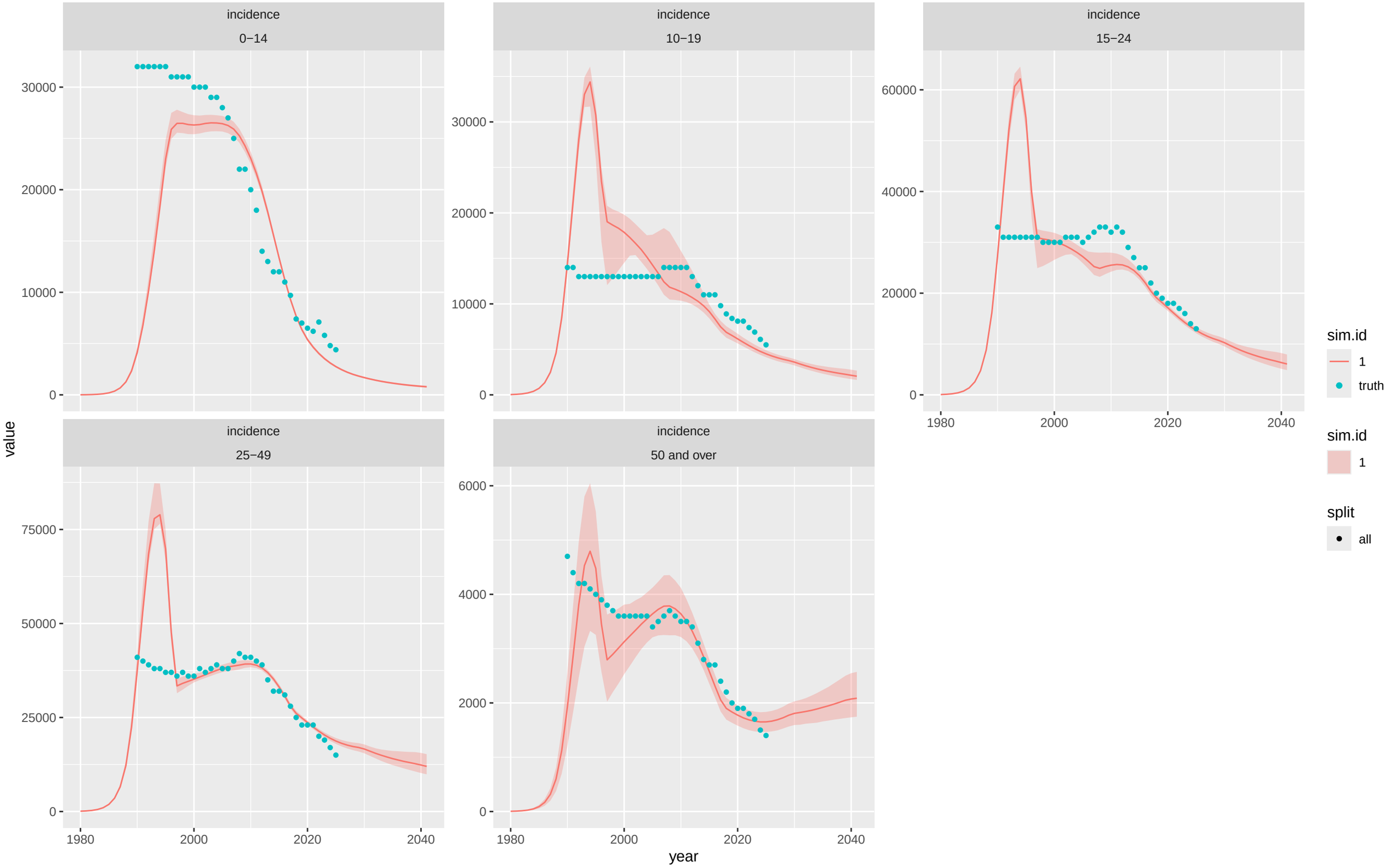
Uganda step 2: hiv.mortality.by.age.sex (one color per chain)



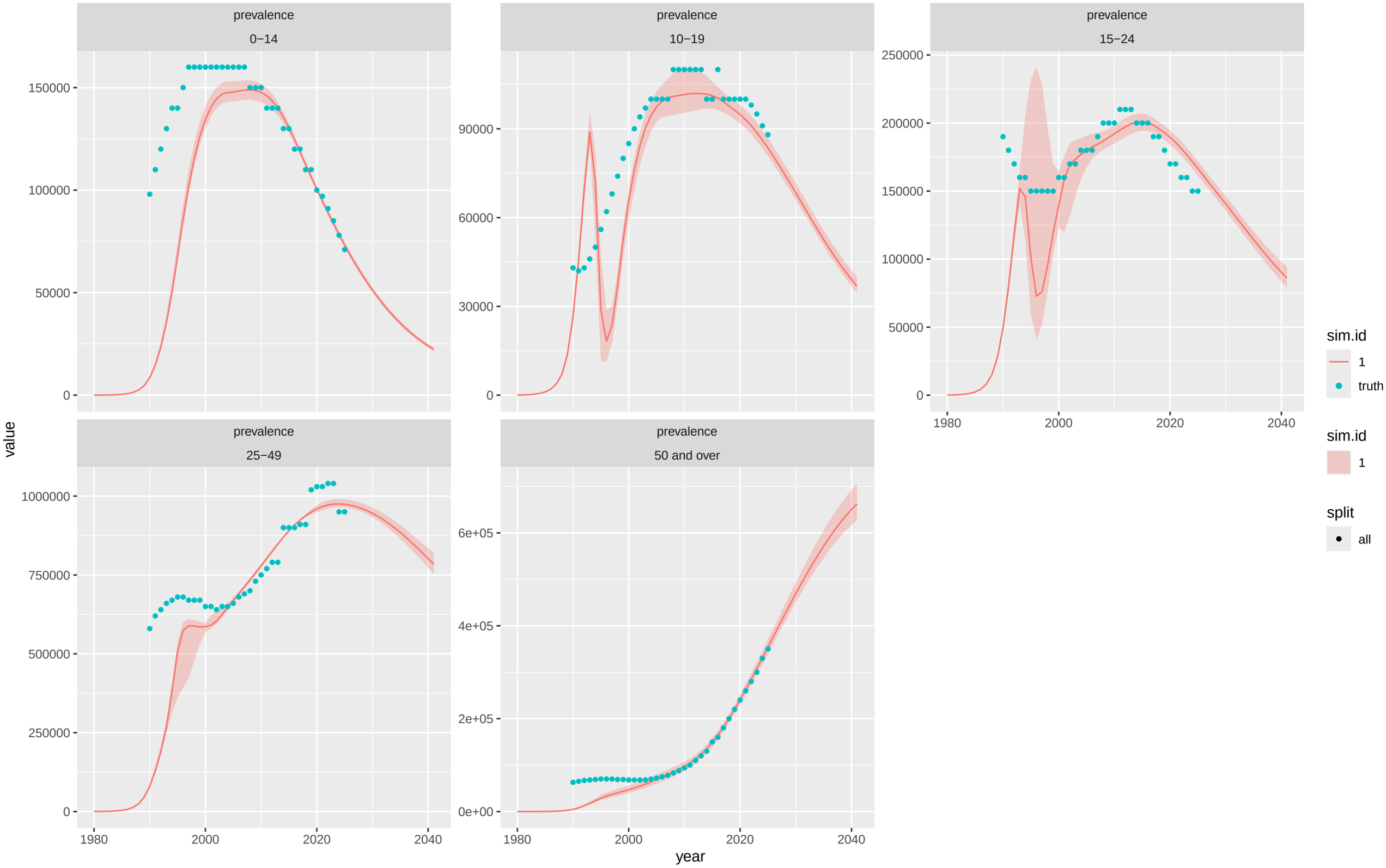
Uganda: fit (merged Step 2 simset)



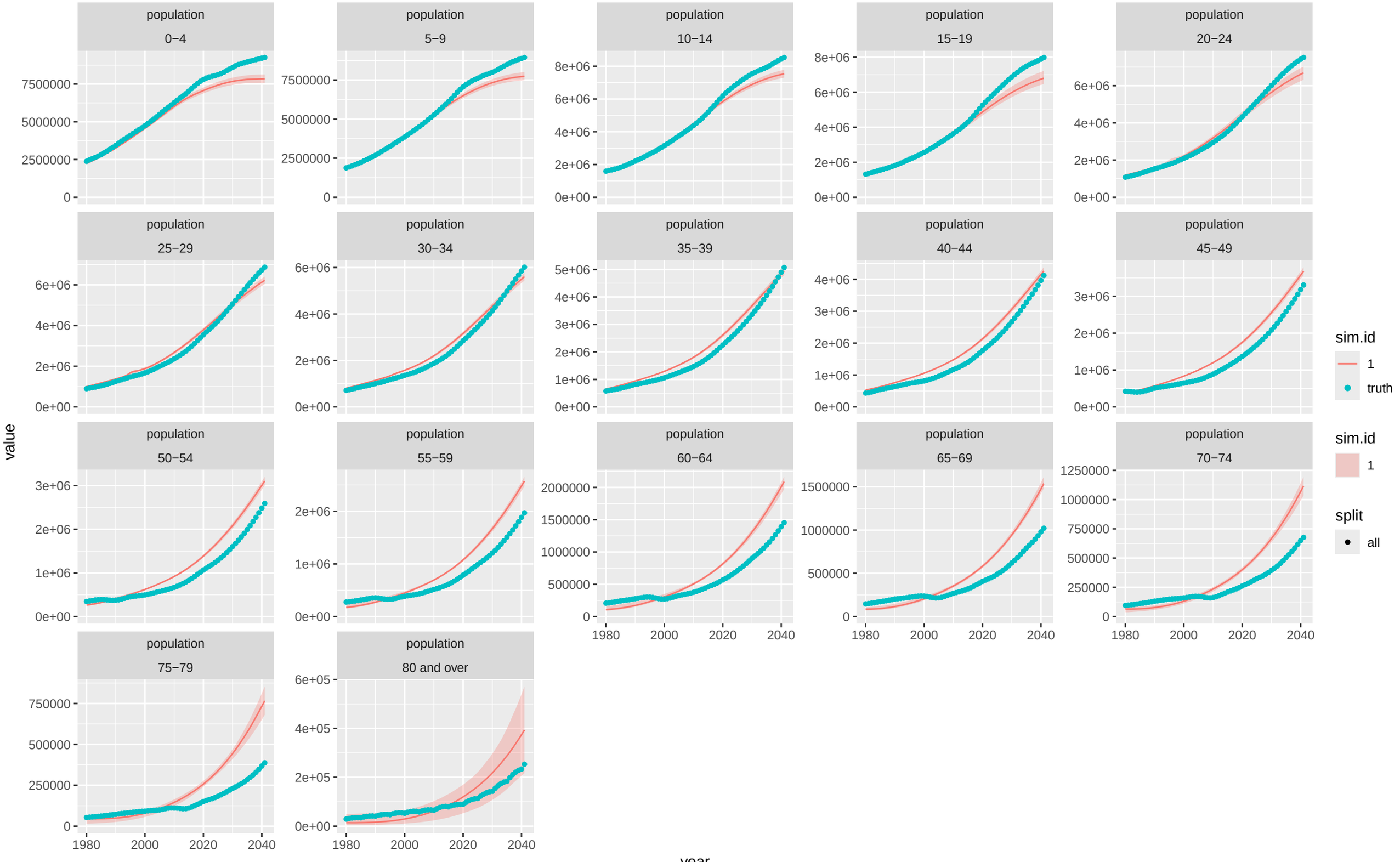
Uganda: incidence.by.age (merged Step 2 simset)



Uganda: prevalence.by.age (merged Step 2 simset)



Uganda: population.by.age (merged Step 2 simset)



Uganda: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

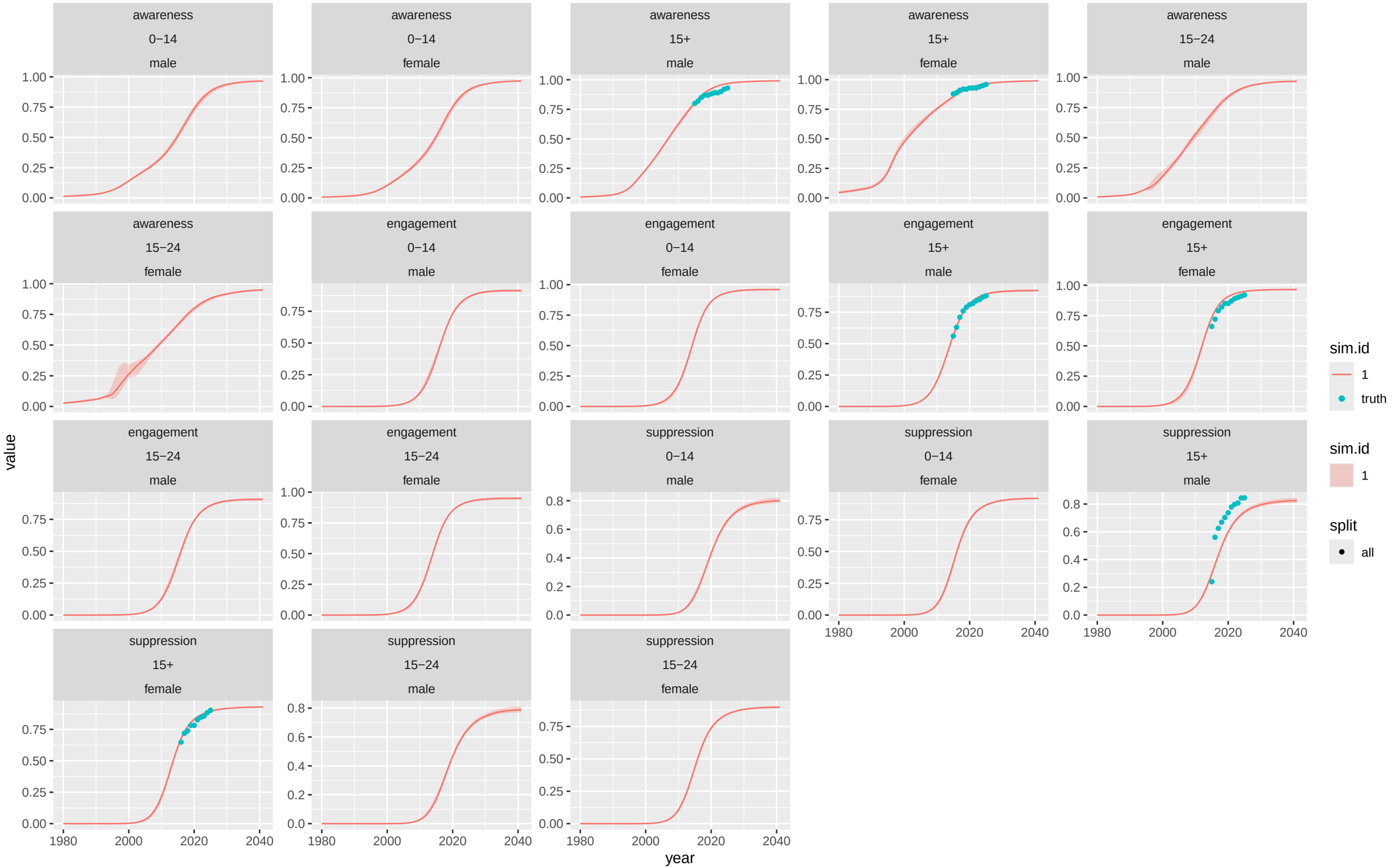
sim.id

1

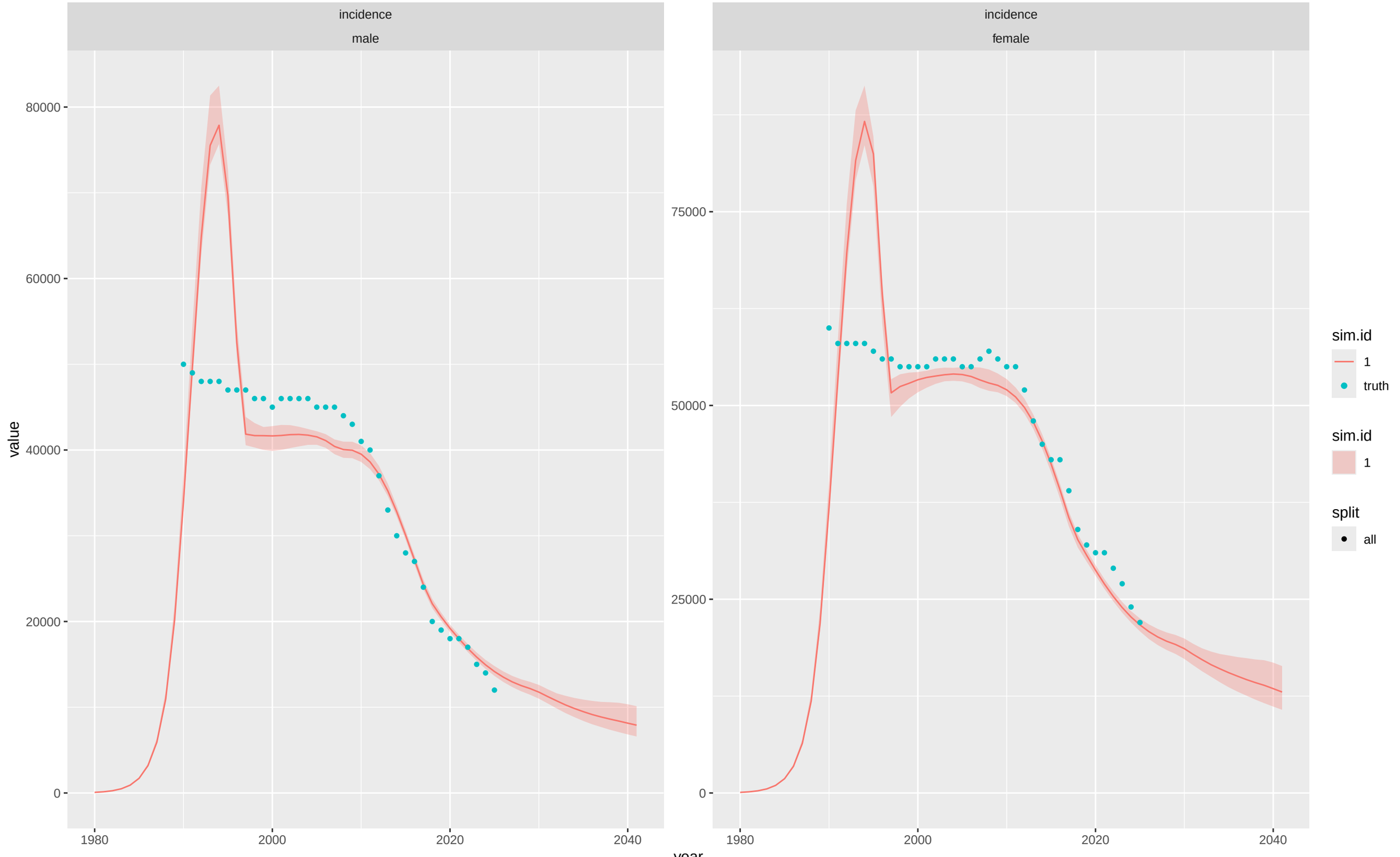
split

all

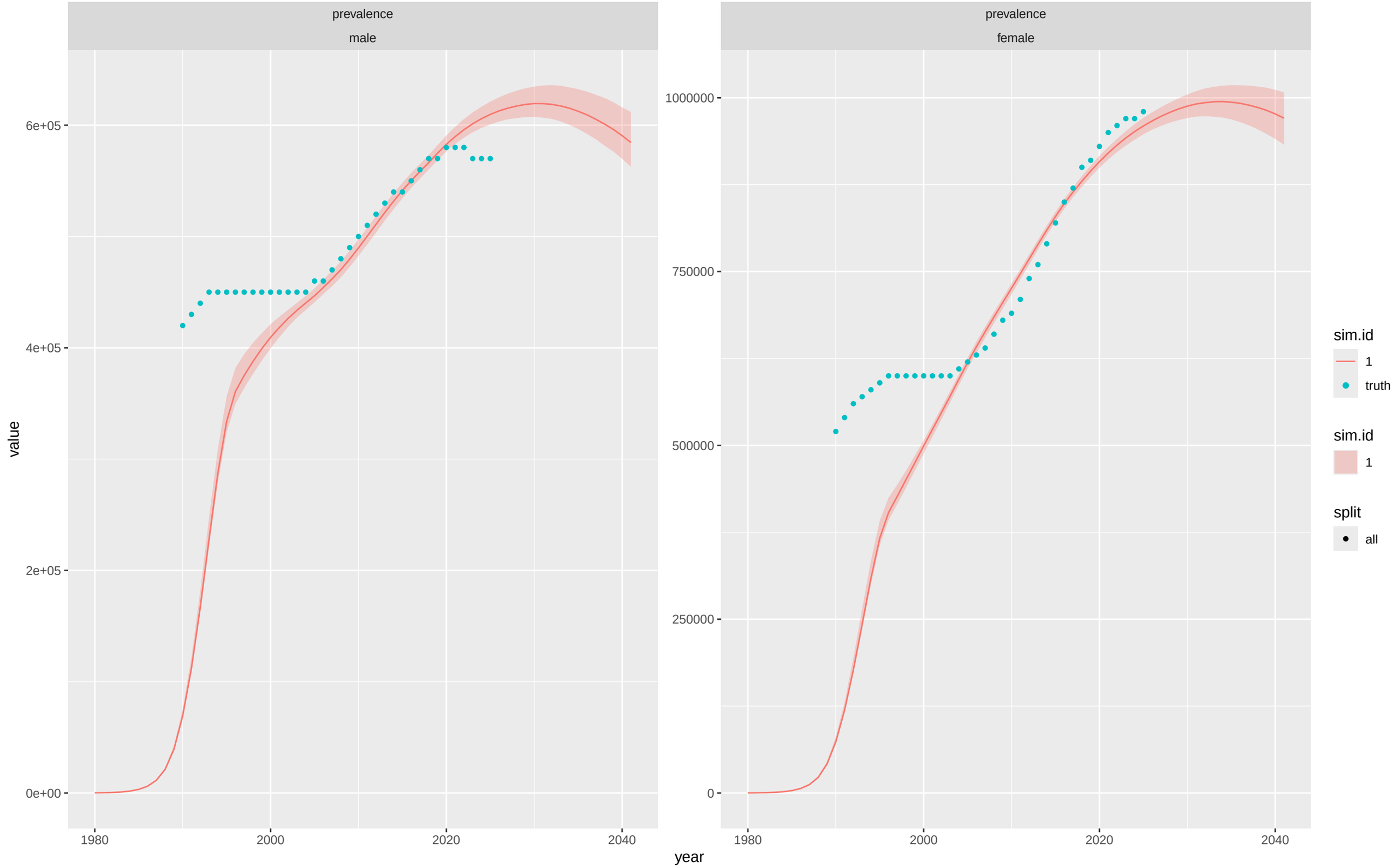
Uganda: cascade.by.age.sex (merged Step 2 simset)



Uganda: incidence.by.sex (merged Step 2 simset)



Uganda: prevalence.by.sex (merged Step 2 simset)



Uganda: hiv.mortality.by.sex (merged Step 2 simset)



Uganda: incidence.by.age.sex (merged Step 2 simset)

incidence 0−14 male
incidence 0−14 female
incidence 10−19 male
incidence 10−19 female
incidence 15−24 male
incidence 15−24 female
incidence 25−49 male
incidence 25−49 female
incidence 50 and over male
incidence 50 and over female

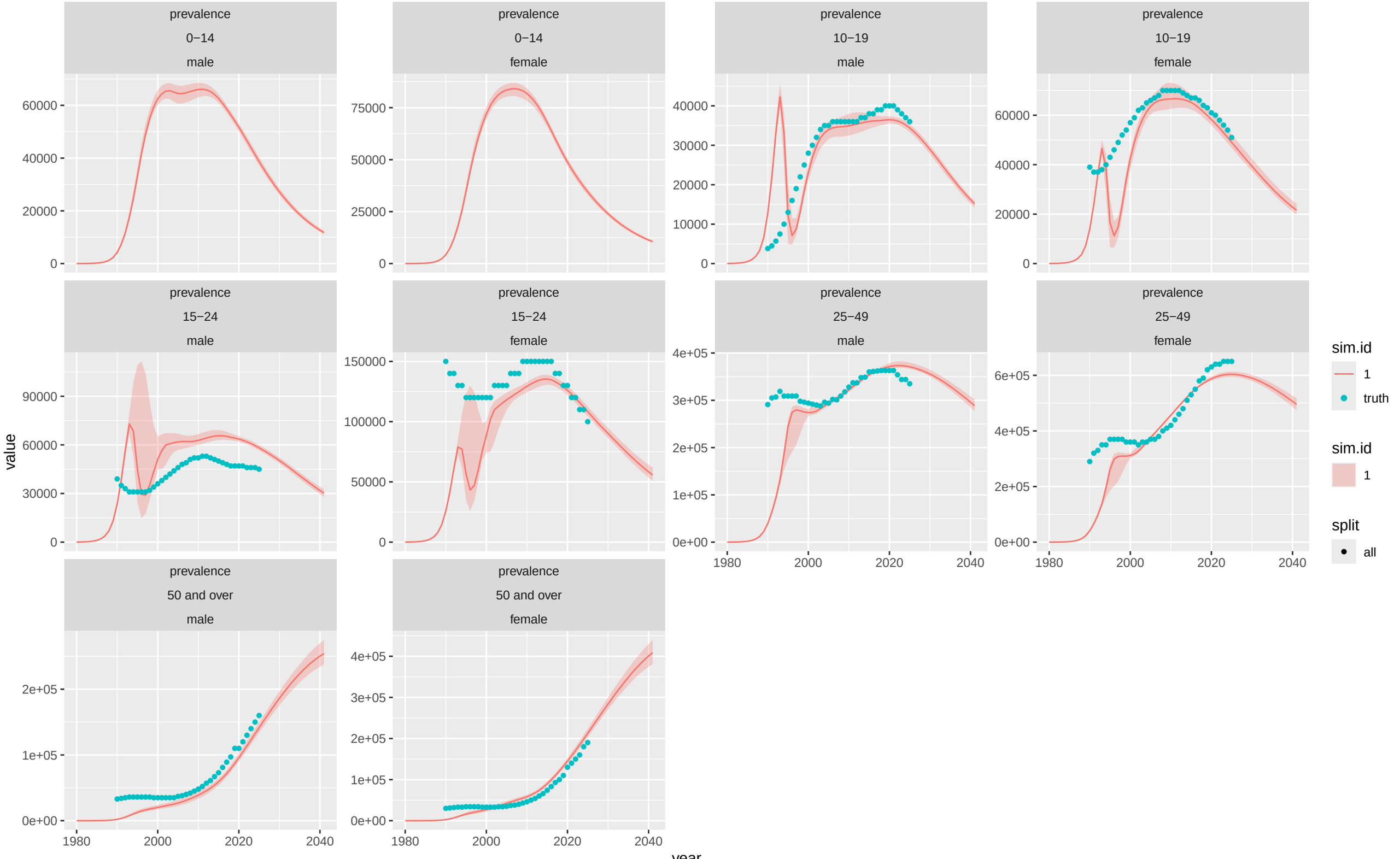
value
year

sim.id
1
truth

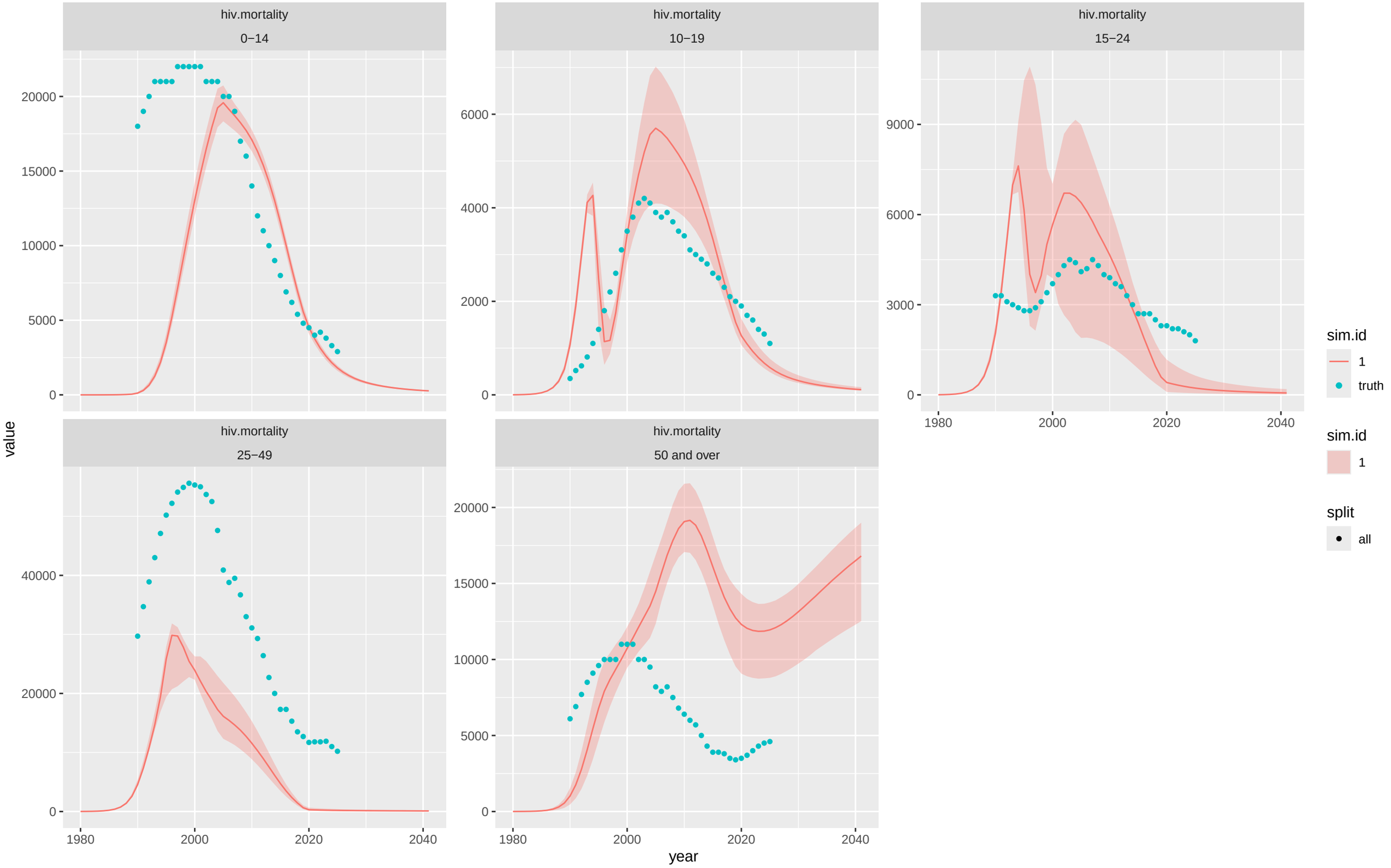
sim.id
1

split
all

Uganda: prevalence.by.age.sex (merged Step 2 simset)



Uganda: hiv.mortality.by.age (merged Step 2 simset)



Uganda: hiv.mortality.by.age.sex (merged Step 2 simset)

